

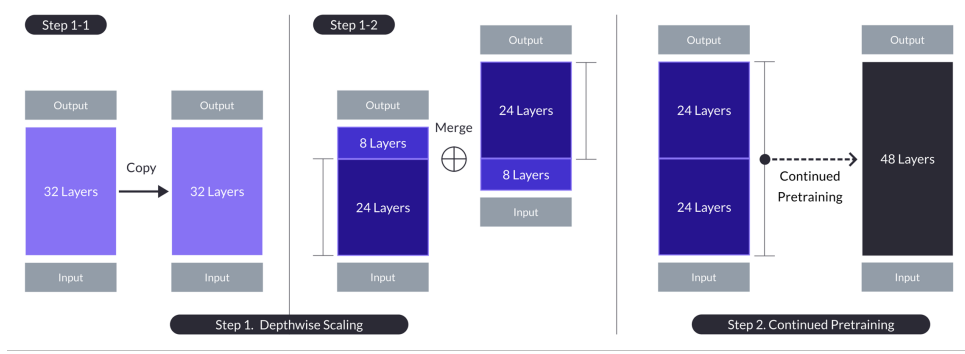


Figure 1: Depth up-scaling for the case with $n = 32$, $s = 48$, and $m = 8$. Depth up-scaling is achieved through a dual-stage process of depthwise scaling followed by continued pretraining.

for wider access and application of these models by researchers and developers globally.

2 Depth Up-Scaling

To efficiently scale-up LLMs, we aim to utilize pre-trained weights of base models to scale up to larger LLMs (Komatsuzaki et al., 2022). While existing methods such as Komatsuzaki et al. (2022) use MoE (Shazeer et al., 2017) to scale-up the model architecture, we opt for a different depthwise scaling strategy inspired by Tan and Le (2019). We then continually pretrain the scaled model as just scaling the model without further pretraining degrades the performance.

Base model. Any n -layer transformer architecture can be used but we select the 32-layer Llama 2 architecture as our base model. We initialize the Llama 2 architecture with pretrained weights from Mistral 7B, as it is one of the top performers compatible with the Llama 2 architecture. By adopting the Llama 2 architecture for our base model, we aim to leverage the vast pool of community resources while introducing novel modifications to further enhance its capabilities.

Depthwise scaling. From the base model with n layers, we set the target layer count s for the scaled model, which is largely dictated by the available hardware.

With the above, the depthwise scaling process is as follows. The base model with n layers is duplicated for subsequent modification. Then, we remove the final m layers from the original model and the initial m layers from its duplicate, thus forming two distinct models with $n - m$ layers. These two models are concatenated to form a scaled model with $s = 2 \cdot (n - m)$ layers. Note that $n = 32$ from our base model and we set $s = 48$ considering

our hardware constraints and the efficiency of the scaled model, *i.e.*, fitting between 7 and 13 billion parameters. Naturally, this leads to the removal of $m = 8$ layers. The depthwise scaling process with $n = 32$, $s = 48$, and $m = 8$ is depicted in ‘Step 1: Depthwise Scaling’ of Fig. 1.

We note that a method in the community that also scale the model in the same manner² as ‘Step 1: Depthwise Scaling’ of Fig. 1 has been concurrently developed.

Continued pretraining. The performance of the depthwise scaled model initially drops below that of the base LLM. Thus, we additionally apply the continued pretraining step as shown in ‘Step 2: Continued Pretraining’ of Fig. 1. Experimentally, we observe rapid performance recovery of the scaled model during continued pretraining, a phenomenon also observed in Komatsuzaki et al. (2022). We consider that the particular way of depthwise scaling has isolated the heterogeneity in the scaled model which allowed for this fast performance recovery.

Delving deeper into the heterogeneity of the scaled model, a simpler alternative to depthwise scaling could be to just repeat its layers once more, *i.e.*, from n to $2n$ layers. Then, the ‘layer distance’, or the difference in the layer indices in the base model, is only bigger than 1 where layers n and $n + 1$ are connected, *i.e.*, at the seam.

However, this results in maximum layer distance at the seam, which may be too significant of a discrepancy for continued pretraining to quickly resolve. Instead, depthwise scaling sacrifices the $2m$ middle layers, thereby reducing the discrepancy at the seam and making it easier for continued

²<https://huggingface.co/Undi95/Mistral-11B-v0.1>